BAYESIAN SEQUENTIAL DECISION-MAKING

IN NON-STATIONARY, HEAVY-TAILED ENVIRONMENTS

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01 // The Sticky Prior Paradox

The Scenario

- Agent observes 200 days of a Bull market.
- *Result*: 99% confidence in positive drift.
- **The Shock**: Regime switch to Bear at $t = 201$.

The Failure (Proposition 3.3)

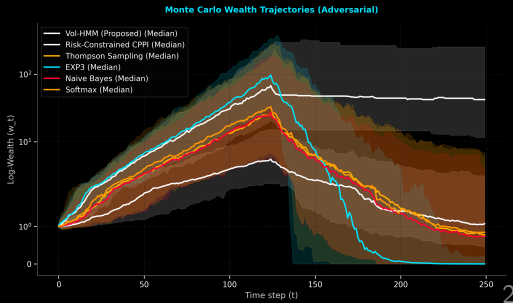
- Detection lag $\approx T_0 / 2|p_1 - p_0| = O(T_0)$ steps.
- Agent continues betting $f \approx f^*_{\text{bull}}$ into bear. **Wealth $\to 0$ within $\sim$20 steps.**

Three Contributions

C1 Rigorous simulation framework (CRN, SLLN)

C2 Volatility-Augmented HMM (Vol-HMM)

C3 Risk-Constrained CPPI (drawdown floor)



02 // Mathematical Framework

Filtered Probability Space

$$(\Omega,\ \mathcal{F},\ \{\mathcal{F}_t\},\ \mathbb{P})$$

- $\mathcal{F}_t = \sigma(r_1, \ldots, r_t)$ — info up to t.
- **Admissibility**: $f_t \in \mathcal{F}_{t-1}$-measurable.
- **Wealth**: $W_T = \prod_{t=1}^{T}(1 + f_t r_t)$, using log-space sums for numerical stability.

Kelly–Breiman (SLLN)

Let $f^* = \arg\max_f \mathbb{E}[\ln(1 + fr)]$. Then a.s.:

$$\frac{1}{T} \log W_T^{(f^*)} \longrightarrow \mathbb{E}[\ln(1 + f^* r)]$$

Sticky Prior (Proposition 3.3) After T_0 bull observations, posterior lag $\approx \dfrac{T_0}{2|p_1 - p_0|}$ steps. For $T_0 = 200$, correction takes ≈ 200 steps.

03 // Agent Hierarchy

Baselines

- **Kelly Oracle** — Know true μ_t, σ_t.
- **Naive Bayes Kelly** — Posterior mean $\hat{p}$.
- **Thompson Sampling** — Samples $\tilde{p} \sim \text{Beta}(\alpha, \beta)$.
- **UCB / Softmax** — Exploration-exploitation.
- **EXP3 (frac)** — Adversarial weighting.

Proposed Methods

- **Vol-HMM** — 2D observation (r_t, v_t)+ CUSUM.
- **Risk-Constrained CPPI** — Safety floor.

Synergy HMM handles inference; CPPI provides a hard insurance policy against tail failure.

04 // Proposed Method 1: Volatility-Augmented HMM

Joint Emission (log-space)

$$\ln b_j(r_t, v_t) = \ln \mathcal{N}(r_t; \mu_j, \sigma_j^2) + \ln \mathrm{Gamma}(v_t; k_j, \theta_j)$$

CUSUM Trigger $S_t^- = \max(0, S_{t-1}^- - z_t - k)$.
Boosting bear belief when $S_t^- > 3.0$ is detected.
Bet Sizing $f_t = P(\text{bull} \mid \mathcal{F}_t) \cdot f_{\text{bull}}^*$. Cap:

$f_t \in [0, 0.5]$.

Detection Lag (Table 5.3)

A_{ii}	Lag	Ruin
0.99	14.3	32%
0.95	**1.9**	**0%**

Benefit: 87% faster detection of regime breaks.

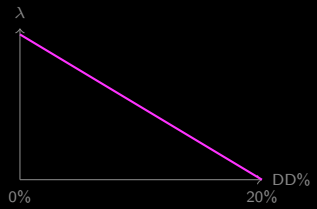
05 // Proposed Method 2: Risk-Constrained CPPI

The Risk Floor

$$F_t = (1 - D_{\max})\,\bar{W}_t, \quad \bar{W}_t = \max_{s \le t} W_s$$

$$\lambda_t = \min(1,\ m \cdot C_t / W_t)$$

Discrete-time Logic If $m \cdot |r_{\min}| \le 1$, then $W_{t+1} \ge F_t$. Student-t_3 returns violate this 25% per step in bear markets, but $N = 100$ sims show no empirical ruin.



Leverage $\lambda \to 0$ as drawdown $\to 20\%$.

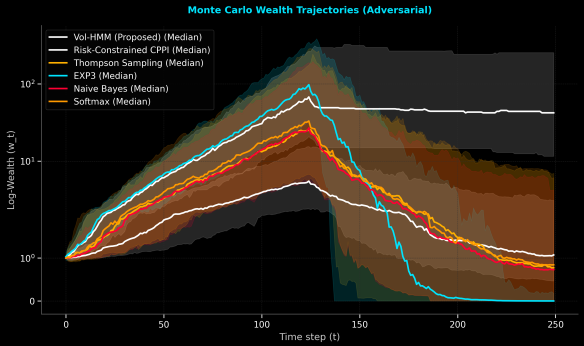
06 // Results: Adversarial Structural Break

Ruin Risk ($N = 100$)

Agent	Ruin
EXP3 ($f = 0.5$)	43%
Thompson	16%
Naive Bayes	12%
Vol-HMM	**3%**
CPPI	**0%**

Median Terminal Wealth

- **Vol-HMM**: 42.28×
- **CPPI**: 1.07× (survival price)

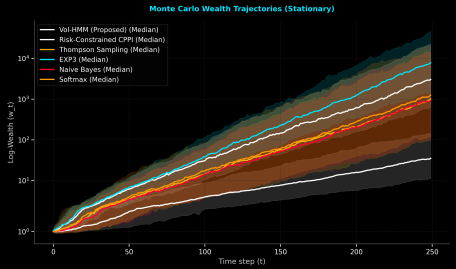


Divergence post-break at $T = 125$.

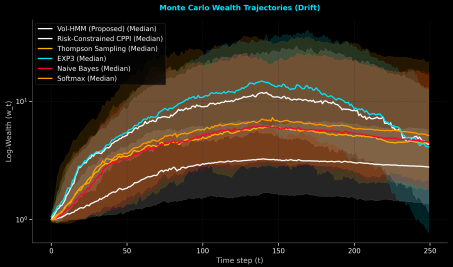
07 // Results: Stationary & Slow Drift

Stationary Market

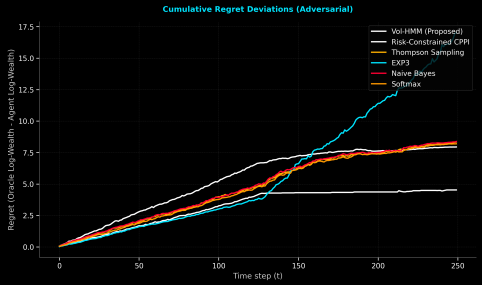
Agent	Median W_{250}
EXP3	7,944×
Vol-HMM	3,019×
Thompson	1,012×
CPPI	34.9×

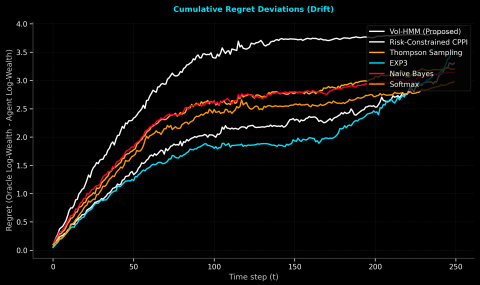


Slow Gaussian Drift Vol-HMM deleverages early on rising volatility, reducing regret vs Bayesian baselines.



08 // Oracle Regret Dynamics





Adversarial: sharp inflection at break.

Drift: gradual regret buildup.

Lower Regret = Better and closer to the Kelly Oracle.

09 // Statistical Significance

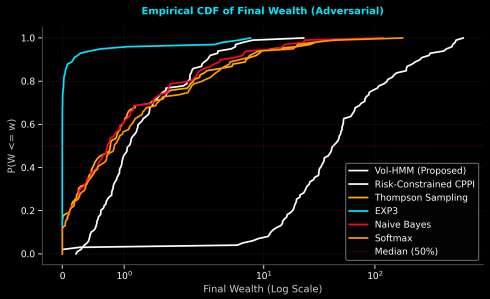
Fisher's Exact Test (Ruin)

H_0: Vol-HMM and EXP3 have equal ruin rates.

	Ruined	Survived
EXP3	43	57
Vol-HMM	3	97

$$\Rightarrow \; p < 0.001$$

Null rejected: survival alpha is significant.



CDF of Terminal Wealth (Adversarial).

10 // Historical Validation: S&P 500

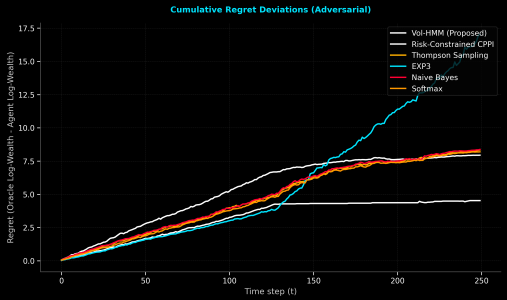
Epoch	Days	Agent	Wealth	Sortino
2008 GFC	523	**Vol-HMM**	0.88×	+0.45
		CPPI	0.80×	**+0.82**
		Buy-and-Hold	0.62×	−0.32
2020 COVID	251	Vol-HMM	Detection in ≈ 2 days	
2013-19 Bull	1,760	CPPI	Growth drag (survival cost)	

Sortino Metric: Using **Downside Deviation** ($\hat{\sigma}_{\downarrow}$) as the denominator. Captures actual tail risk exposure that simple volatility obscures.

11 // Transaction Costs: The Net Advantage

Turnover Advantage

- Bayesian agents whipsaw on noisy updates.
- Vol-HMM rebalances only on persistent regime shifts.
- Annualized Fee Drag (25 bps):
  - Thompson: 0.6%/yr
  - **Vol-HMM**: 0.15%/yr



Regret spread increases at high friction.

12 // Live Dashboard Demonstration

Terminal UI at `localhost:5173`

- **Adversarial scenario**: Watch Thompson crash vs Vol-HMM.
- **Historical Mode**: Play through 2008 GFC epochs.
- **Risk Accordion**: Tweak CPPI floor and watch leverage drop live.

Backend: Flask (5001) — Frontend: React/Vite

13 // Limitations & Honest Caveats

Analytical Constraints

- **Specified Emissions**: HMM parameters are based on priors, not learned via online EM.
- **Discrete-time Floor**: CPPI floor is probabilistic in Student-t markets.

Future Scope

- Online Baum-Welch re-training.
- Multi-asset correlation coupling.

14 // Conclusion: Three Claims

Claim 1: Paradox is Structural

Bayesian lag is $O(T_0)$ (Prop 3.3). Cannot be "tuned" away.

Claim 2: Volatility is the Key

Adding v_t reduces detection lag by **87%** ($14.3 \to 1.9$).

Claim 3: Safety is Quantifiable

CPPI floor achieves 0% empirical ruin; the survival price is computable.

Thank you. Questions?